

CHAPTER 10

SYMBIOTIC RELATIONSHIPS

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

This brief chapter revisits many topics on symbiosis that were introduced earlier in the textbook in a slightly different context. Here, these fascinating phenomena are covered in detail. The chapter is divided into three parts. An initial concise passage, COVERAGE AND DEFINITIONS, reviews the general categories of symbiotic relationships, while the next two sections cover algal–animal (SYMBIOSIS OF ALGAE AND ANIMALS) and animal–animal (SYMBIOSIS AMONG ANIMALS) symbiotic relationships.

Coverage and Definitions

The traditional North American view is used in defining symbiosis (that is, including parasitism as well as all other nonpredator and nonherbivore interactions). European texts treat parasitism and symbiosis as separate, but really coequal, categories.

commensalism	inquilinism	symbionts
facultative associations	mutualism	symbiosis
host	parasitism	symbiotes

I. Symbioses Among Algae and Animals

This section considers in some detail the symbiotic relationships between algae and invertebrates. This examination is accomplished in seven subsections:

- Types and Composition of the Associations
- Origin of the Associations
- Distribution of Algal – Invertebrate Associations
- Modifications Resulting from the Association
- Value of the Association
- Establishment and Transmission of Algal Symbionts
- Symbiosis of the Deep

annelids	Demospongia	platyhelminths
blue-green algae	diatoms	protozoans
calcification	dinoflagellate	radiolarians
<i>Cassiopeia</i>	echinoderms	RNA sequencing
chloroplast symbiosis	echiurans	senile zooxanthellae
Chrysophyceans	endobiotic	sponges
<i>Cliona</i>	epibiotic	syncytium
cnidarian	epizoites	translocation
<i>Convoluta</i>	foraminiferans	transmission
cryptic symbiosis	glycerol	<i>Tridacna</i>
cryptophyceans	haptophyceans	zoochlorellae
ctenophores	jellyfish	zooxanthellae
cyanellae	mollusks	
cyanobacteria	norm of the reaction	

1. Either entire algal cells are symbionts within an invertebrate (perhaps holosymbionts?) or the chloroplasts may function as the symbionts (chloroplast symbiosis). At this point, instructors may wish to discuss the theory of bacterial symbiosis of eukaryotic cells: mitochondria and chloroplasts.
2. Algal symbionts may be grouped into three categories: zooxanthellae (mainly dinoflagellates), zoochlorellae (chlorophytes), and cyanellae (cyanobacteria; blue-green bacteria = blue-green

algae). However, this grouping does not have taxonomic significance. The zooxanthellae comprise dinoflagellates and diatoms, among others.

3. Symbionts may be located within cellular vacuoles or in intercellular spaces, but they are usually restricted to specific regions of the animal. Many invertebrate phyla are involved in such arrangements, including protozoans, sponges, cnidarians, free-living flatworms, mollusks, and echiurans. Some protozoans initially thought to be photosynthetic are now believed to be examples of cryptic symbiosis—a condition in which the algal cell has become indistinguishable from its host.
4. Significant morphological, physiological, and behavioral changes take place in both partners of the symbiosis. Some of these changes can be seen if the algal symbiont is cultured separately from its invertebrate host and then the free-living and symbiotic forms are compared. (See Figure 10.3 in the textbook.) These changes fall into the widespread phenomenon of the norm of the reaction (also referred to as the reaction norm); the organisms have a genotype that is plastic enough to allow for a certain variety of form, depending on the specific conditions of growth. Vermeij (1987) describes the norm of reaction as “a very widespread phenomenon . . . [in which] a change in conditions causes a predictable alteration in the phenotype. Norms of reaction are the rule in the dependence of reproductive characteristics on environmentally controlled growth rates . . .”

Vermeij, G. 1987. *Evolution and escalation: an ecological history of life*. Princeton, NJ: Princeton University Press.

5. Algal symbionts are found in the outer frothy layer of all epipelagic radiolarians. Modifications to the symbiotic condition in the invertebrate host vary from minor (as in sponges) to dramatic (as in tridacnid clams). The extent of these evolutionary modifications is reviewed in the textbook.
6. It appears intuitively obvious that these relationships provide a significant advantage for the symbionts. The textbook reviews several arguments for and the evidence that supports the concept of an adaptive significance of symbiotic relationships. It might be useful to initiate a discussion of these topics so that students consider the concept of coevolution, in which the evolution of morphological, physiological, and behavioral characteristics in the two symbionts is explored.
7. Passage of the algal symbiont from generation to generation depends on the invertebrate in question. For some species, a completely new infection (infestation) must be established. In other phyla, the egg or larva is seeded with the appropriate symbionts.

II. Symbioses Among Animals

This section provides a fairly comprehensive study of the positive aspects of symbiotic relationships among animals. This examination is accomplished in five subsections:

- Types of Associations
- Origin of the Association
- Modifications Due to the Association
- Value of the Associations
- Luminescent Bacteria

acclimatization	ectocommensalism	pearlfish
anemone	endozoites	photobacterium
annelids	entoprocts	<i>Physalia</i>
arthropods	epizoites	pinnothereid
bioluminescence	guest	Portuguese man-of-war
camouflage	hermit crab	protozoans (protists)
<i>Chaetopterus</i>	host	remora
cleaning behavior	luminescent bacteria	scale worms
cleaning station	mollusks	Texas longhorns
commensal	mutualism	thigmotaxis
copepods	nematocysts	tunicin
crustaceans	nemertine	turbellarians
echiurans	pea crab	whale barnacle

1. Commensalism among marine animals occurs both internally (endozoites) and externally (epizoites). Eliminated from further discussion are those fortuitous associations of larvae settling on the outer surface of another animal. However, many groups, including protozoans, cnidarians, entoprocts, annelids, arthropods, and mollusks, are involved in true epizoitic relationships. Numerous endozoitic and epizoitic associations (relationships) are reviewed in the text.
2. Once again, anatomical, physiological, and behavioral alterations are present in both symbionts, with the fewest number of changes being evident in epizoitic relationships and the greatest number in endozoitic ones. Many commensals exhibit thigmotaxis.
3. The phenomenon of cleaning symbiosis is an interesting example of the prisoner's dilemma, or the ultimate game.

Nowak, M.A., K.M. Page, and K. Sigmund. 2000. Fairness versus reason in the ultimate game. *Science* 289: 1773–1775.

4. The use of the terms “choice,” “choose,” “strategy,” “select,” “habitat selection,” and “recognize,” among others, is rather anthropomorphic and more than a wee bit teleological. Unfortunately, it is difficult to discuss the concepts of host (that is, habitat) recognition without using these phrases. Instructors might remind their students that researchers use these terms as a matter of convenience. The terms are not meant to convey the idea that organisms cognitively compare one habitat with another and then select the better one.
5. The value of the symbiotic relationship in animal–animal associations varies greatly. For some organisms, it is just a matter of having a safe haven, while for others, food is an added benefit. In cleaning symbioses, both symbionts gain: One obtains food, and the other experiences reduced parasite infestation.
6. Bioluminescent bacteria and their hosts (for example, fish and squid) are a final subject for this chapter. See also Chapter 4 for more information on this topic.

ENRICHMENT EXERCISES

Symbiotic Relationships

Gilbert and Allen (1973a and b) have shown that, in a shallow pond in New Hampshire, the freshwater sponge *Spongilla lacustris*, together with its symbionts, acts as a net primary producer. Their work may be repeated by instructors at landlocked institutions. Gilbert and Allen found that *Spongilla* is capable of achieving rates of approximately 31 mg O₂/g ash-free dry-weight sponge per hour. Mean chlorophyll-*a* concentrations were in the range of 0.3–1.0 µg/mg ash-free dry-weight sponge. Gilbert and Allen's techniques are simple and can be repeated by students on freshwater or marine sponges possessing symbiotic algae.

Pigment Analysis

Sponges were sectioned and their wet weights determined. These pieces were then ground by hand, using a ground-glass homogenizer, with basic, aqueous acetone (1:9 water:acetone, with ≈ 2 ml/L 30% NH₄OH) as the solvent. Weights of these sections were correlated with those of similar pieces that were treated to determine their wet, dry, and ash weights.

Primary Productivity (Oxygen Technique)

Sections of sponge were suspended by thread in light and dark 250-ml B.O.D. and then incubated for 2–3 hours at the depth from which they were collected. Light and dark bottles without sponge sections also were incubated to determine phytoplankton productivity.

Davy, S.K., et al. 2002. Ammonium excretion by a symbiotic sponge supplies the nitrogen requirements of its rhodophyte partner. *J. Exp. Bio.* 205: 3505–3512.

Gilbert, J.J., and H.L. Allen. 1973. Studies on the physiology of the green freshwater sponge, *Spongilla lacustris*: primary productivity, organic matter, and chlorophyll content. *Verh. Int. Ver. Limnol.* 18: 1413–1420.

Gilbert, J.J., and H.L. Allen. 1973. Chlorophyll and primary productivity of some green, freshwater sponges. *Int. Revue Ges. Hydrobiol.* 58: 633–658.

Symbiosis and Hydrothermal Vents

The symbiotic relationship between certain animals inhabiting hydrothermal vents or the deep-sea floor (see Chapter 4) and bacteria has been the source of much interesting research. For a start on this topic, consult the following references:

- Cavanaugh, C.M. 1994. Microbial symbiosis: patterns of diversity in the marine environment. *Am. Zool.* 34: 79–90.
- Cavanaugh, C.M., et al. 1981. Prokaryotic cells in the hydrothermal vent tube worm *Riftia pachyptila* Jones: possible chemoautotrophic symbionts. *Science* 213: 340–341.
- Cronan, D.S. (ed.). 2000. *Handbook of marine mineral deposits*. Boca Raton, FL: CRC Press.
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- Lutz, R.A. 1991. The biology of deep-sea vents and seeps. *Oceanus* 34(4): 75–83.
- Perkins, S. 2003. Seafloor system has been active for ages. *Science News* 164: 52–53.
- Roberts, D., et al. 1991. Procaryotes on the tentacles of deep-sea holothurians: a novel form of dietary supplementation. *Limnol. Oceanogr.* 36: 1447–1452.
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- Southward, A.J., and E.C. Southward. 1987. “Pogonophora.” Pages 201–228 in Pandian, T.J., and Vernberg, F.J. (eds.). *Animal energetics*. Volume 2. New York: Academic Press.
- Southward, A.J., et al. 1986. Chemoautotrophic function of bacterial symbionts in small Pogonophora. *J. Mar. Biol. Assoc. U.K.* 66: 415–437.
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- Terwillier, R.C., and N.B. Terwilliger. 1985. Respiratory proteins of hydrothermal vent animals. *Biol. Soc. Wash. Bull.* 6: 273–287.
- Tunnicliffe, V. 1992. Hydrothermal-vent communities of the deep sea. *Am. Sci.* 80: 336–349.
- Windoffer, R., and O. Giere. 1997. Symbiosis of the hydrothermal vent gastropod *Ifremeria nautiliei* (Provannidae) with endobacteria—structural analyses and ecological considerations. *Biol. Bull.* 193: 381–392.

Coral Bleaching

Because it is on occasion featured in the popular news, coral bleaching is another topic of much interest to students. For a start on this topic, consult the following references:

- Aronson, R.B., W.F. Precht, and I.G. Macintyre. 2000. Coral bleach-out in Belize. *Nature* 405: 36.
- Aronson, R.B., et al. 2002. The 1998 bleaching event and its aftermath on a coral reef in Belize. *Mar. Bio.* 141: 435–448.
- Brown, B.E., R.P. Dunne, and M.S. Goodson. 2000. Bleaching patterns in reef corals. *Nature* 404: 142–143.
- Buddemeier, R.W., and D.G. Fautin. 1993. Coral bleaching as an adaptive mechanism. *BioScience* 43(5): 320–326.
- Cook, C.B., et al. 1990. Elevated temperatures and bleaching on a high latitude coral reef: the 1988 Bermuda event. *Coral-Reefs* 9(1): 45–49.
- Gates, R.D. 1990. Seawater temperature and sublethal coral bleaching in Jamaica. *Coral-Reefs* 8(4): 193–197.
- Glynn, P.W. 1993. Coral reef bleaching: ecological perspectives. *Coral-Reefs* 12(1): 1–17.

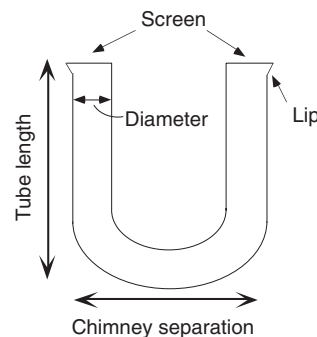
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- Hardy, J.T., et al. 1992. Remote detection of coral “bleaching” using pulsed-laser fluorescence spectroscopy. *Mar. Ecol. Prog. Ser.* 88(2–3): 247–255.
- Hughes, T.P., et al. 2003. Climate change, human impacts, and the resilience of coral reefs. *Science* 301: 929–934.
- Jokiel, P.L., and S.L. Coles. 1990. Response of Hawaiian and other Indo-Pacific reef corals to elevated temperature. *Coral-Reefs* 8(4): 155–162.
- Kleppel, G.S., R.E. Dodge, and C.J. Reese. 1989. Changes in pigmentation associated with the bleaching of stony corals. *Limnol. Oceanogr.* 34: 1331–1335.
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- Porter, J.W., et al. 1989. Bleaching in reef corals: physiological and stable isotopic responses. *Proc. Natl. Acad. Sci. USA* 86(23): 9342–9346.
- Sebens, K.P. 1994. Biodiversity of coral reefs: what are we losing and why? *Am. Zool.* 34(1): 115–133.
- Szmant, A.M., and N.J. Gassman. 1990. The effects of prolonged “bleaching” on the tissue biomass and reproduction of the reef coral *Montastrea annularis*. *Coral-Reefs* 8(4): 217–224.
- West, J.M., and R.V. Salm. 2003. Resistance and resilience to coral bleaching: implications for coral reef conservation and management. *Conserv. Bio.* 17: 956–968.
- Wilkinson, C., O. Linden, and H. Cesar. 1999. Ecological and socioeconomic impacts of 1998 coral mortality in the Indian Ocean: an ENSO impact and a warning of future change? *Ambio* 28(2): 188–196.

Other Invertebrate Relationships

1. *Chaetopterus variopedatus* (Annelida, Polychaeta, the parchment tube worm) and *Pinnixa chaetopterana* (a pinnotherid crab):

If you have the opportunity, have your students observe this polychaete in U-shaped glass tubes. The ones I use are 2–3 cm in diameter and 15–20 cm in length, with a bend that separates each chimney by 10–15 cm. (See the accompanying figure.) Keep a trickle of seawater flowing through the tube, but place some netting over both ends of the tube, or else the worm might slide out of the tube and down the drain. The worm has a spatulate region at the anterior end that acts as a kind of holdfast organ, but if the flow of water is too fast or if conditions are not right, the worm may give up.

The worm is bioluminescent, making a weak blue light when prodded with a sharp probe. The light spreads out from the point of stimulation. With this system, one may observe a variety of activities, including the following: feeding activities of both the worm and its commensal crab; worm–crab behavioral interactions; bioluminescence from the worm. Finally, if left alone, the worm will begin to lay down tube material, which will cause the sides of the glass to become opaque with the new parchment.



2. *Limulus polyphemus* (Arthropoda, Chelicerata, the horseshoe crab) and *Bdelloura candida* (Platyhelminthes, Turbellaria, the *Limulus* leech):

Adult specimens of the horseshoe crab almost always have *Bdelloura* on their legs. (At least, the ones on the Gulf coast always do, as far as I can tell.) If adult specimens of this chelicerate are available, have your students examine the number and distribution of the flatworms. Are they evenly distributed, or do they have a tendency to be concentrated in one location? Do males and females have the same relative number of *Bdelloura*? Is the size of *Limulus* a factor in the number of *Bdelloura* present? You might also have your students explore the question of whether the flatworm is parasitic, phoretic, or commensal.

At the same time, one could look at the distribution of barnacles on *Limulus*. Is there a pattern to their positioning? Do males and females have the same relative number of barnacles? (Be sure to account for the size of the arthropod.)

Anon. 2002. Chemical from controversial horseshoe crab is vital to human health. *Ascribe Higher Education News Service*. Jan. 31, 2002.

Chang, H.S., and D.A. Sack. 2001. Detection of anti-lipopolysaccharide antibodies to *Vibrio cholerae* O1 and O139 using a novel microtiter limulus amebocyte lysate (LAL) assay. *Clin. Chim. Acta* 312: 49–55.

Harrington, J.M., and P.B. Armstrong. 1999. A cuticular secretion of the horseshoe crab, *Limulus polyphemus*: a potential anti-fouling agent. *Biol. Bull.* 197: 274–275.

Maury, L. 2002. Blood of the ancient mariner. *The Ecologist* 32: 27–28.

Penn, D., and H.J. Brockmann. 1994. Nest-site selection in the horseshoe crab, *Limulus polyphemus*. *Biol. Bull.* 187: 373–384.

Wainwright, N.R., and T. Novitsky. 1997. Endotoxin binding and neutralizing protein and uses thereof. *Biotechnology Advances*, Annual, 15: 499–500.

Widener, J.W., and R.B. Barlow. 1999. Decline of a horseshoe crab population on Cape Cod. *Biol. Bull.* 197: 300–302.